

August 20th, 2025

To whom it may concern,

I am writing this letter to assist you in understanding Hedong Ren's performance in the Introduction to Multi-Armed Bandit Algorithms, which began with one week of online instruction consisting of two lectures, and two weeks of onsite daily instruction consisting of eight lectures in Beijing, China, in the summer of 2025. Being the leading instructor of this program and witnessing the progress, I've included a program summary for your reference to provide you with an idea of what the student has accomplished in this program.

This is an introductory course on multi-armed bandits, which provides a sequential decision-making framework under uncertainty and has broad applications in recommendation systems, dynamic pricing, clinical trials, financial investments, etc. We will cover the classical multi-armed bandit model and its applications, several widely used algorithms proposed for its solution including the Explore-Then-Commit (ETC), Upper Confidence Bound (UCB) and Thompson Sampling (TS) Algorithms, performance analysis of these algorithms, and conclude the lectures with the recent work of the instructor on correlated and structured bandits.

Overall, Hedong Ren's performance in this program was excellent.

Hedong is a valuable team player. Not only did he display his ability to effectively discuss topics covered in the class, but Hedong also demonstrated sufficient open-mindedness to understand others' opinions. For most of the time, he contributed to the discussion by providing interesting examples and opinions. Hedong showed excellent engagement throughout the course. He was quick to respond to questions and frequently contributed insightful points during lectures and breakout discussions. His professional attitude and active participation helped build a collaborative learning atmosphere.

Hedong's assignment performance was excellent. He submitted all the assignments on time and his homework questions were consistent of great quality. Hedong submitted mathematically rigorous and well-annotated assignments. He was particularly strong in optimizing the computational logic behind simulation-based tasks and always included insightful interpretations of numerical outcomes. His consistency and attention to detail stood out.

In the end of the program, Hedong's group prepared their presentation based on the summary of the course content. Group 2 conducted a technically sound project exploring multiple multi-armed bandit algorithms using the MovieLens recommendation dataset. The group demonstrated a solid understanding of the subject matter, with each member contributing a structured and meaningful segment to the presentation. The algorithmic comparisons were well executed, backed by thoughtful simulations and relevant commentary. This project reflected the team's collaborative spirit,

commitment to preparation, and ability to apply theoretical concepts in a practical context. Hedong demonstrated strong technical and communication skills throughout his segment of the project. His explanation of the Thompson Sampling algorithm and subsequent discussion of UCB were both clear and comprehensive. He presented dynamic visualizations that effectively illustrated posterior updates, regret accumulation, and arm selection trends, highlighting his deep grasp of the material. His confident interpretation of exploration coefficients in UCB showed critical thinking and analytical precision. Hedong's professionalism, fluent delivery, and insightful commentary were key strengths that elevated the quality of the group's final presentation.

I hope you will find this letter useful in evaluating Hedong Ren's academic performance as a reference.

Sincerely,



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